

KUSONG, Korea, Democratic People's Republic of

WMO: 470370

Lat: 39.983N

Lon: 125.250E

Elev: 325

StdP: 14.52

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.1	(c) 5.6	(d) -10.6	(e) 3.1	(f) 8.3	(g) -7.4	(h) 3.7	(i) 10.4	(j) 16.1	(k) 14.5	(l) 14.1	(m) 15.1	(n) 2.4	(o) 320	(p) 0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.5	(c) 88.9	(d) 76.9	(e) 86.5	(f) 75.0	(g) 84.3	(h) 74.1	(i) 79.7	(j) 85.8	(k) 77.9	(l) 83.3	(m) 76.5	(n) 81.1	(o) 4.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	147.4	83.5	76.4	139.2	81.0	75.1	133.3	79.5	43.7	85.9	41.8	83.6	40.3	81.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	11.1	9.4	-3.1	93.4	4.3	2.1	-6.2	95.0	-8.8	96.2	-11.2	97.4	-14.3	98.9
			-3.6	82.0	4.3	2.6	-6.7	83.8	-9.2	85.3	-11.7	86.8	-14.8	88.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	20.4	25.7	37.1	50.1	61.9	70.5	75.9	76.3	67.3	54.5	37.7	23.4	
	DBStd	20.83	7.09	7.22	6.61	6.64	5.30	3.79	3.66	3.98	4.84	6.58	8.68	7.75	
	HDD50	3312	918	679	402	78	1	0	0	0	0	32	376	826	
	HDD65	6390	1383	1099	865	446	124	4	0	1	31	327	819	1291	
	CDD50	3387	0	0	2	82	370	615	803	815	519	173	8	0	
	CDD65	989	0	0	0	0	28	169	338	351	100	3	0	0	
	CDH74	6877	0	0	0	15	325	1104	2231	2576	599	27	0	0	
	CDH80	1833	0	0	0	0	56	263	634	793	85	2	0	0	
Wind	WSAvg	2.8	3.2	3.3	3.2	3.3	3.0	2.6	2.4	2.2	2.2	2.5	2.8	3.2	
Precipitation	PrecAvg	47.9	0.3	0.8	1.0	2.2	3.7	4.9	14.8	11.3	4.4	2.1	1.9	0.7	
	PrecMax	81.5	1.1	3.7	3.2	4.5	8.3	8.8	41.6	27.7	12.0	4.9	5.3	2.2	
	PrecMin	33.4	0.0	0.0	0.0	0.1	0.4	1.7	4.9	2.7	0.7	0.2	0.3	0.1	
	PrecStd	12.3	0.3	0.9	0.6	1.2	1.9	2.2	7.2	6.4	3.1	1.3	1.5	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	48.2	62.1	76.2	85.3	88.8	93.2	92.6	85.7	77.4	62.8	44.3	
		MCWB	35.7	38.9	50.8	59.2	66.6	71.6	80.1	80.1	72.7	64.3	53.6	39.0	
	2%	DB	36.8	43.6	56.4	71.3	80.9	85.8	88.9	89.2	82.4	72.6	56.8	40.3	
		MCWB	31.7	36.6	47.0	56.9	64.3	70.6	77.7	78.6	70.8	61.6	50.6	36.5	
	5%	DB	34.1	39.9	52.5	67.3	77.4	83.0	86.0	86.8	80.1	69.1	53.6	37.1	
		MCWB	30.3	33.9	44.1	55.1	62.7	69.7	76.3	76.2	68.8	59.3	47.5	33.6	
	10%	DB	31.4	36.7	48.6	63.3	74.2	80.5	83.6	84.6	77.6	65.9	50.3	34.5	
		MCWB	27.6	31.6	41.2	51.9	61.1	68.7	75.6	75.2	67.4	57.1	45.8	31.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	41.7	51.8	61.9	69.3	75.3	82.8	83.2	75.6	66.5	55.2	41.1
			MCDB	40.0	46.1	59.4	71.4	81.0	84.8	90.4	89.9	82.0	74.1	60.4	42.7
2%		WB	33.0	37.5	48.4	58.6	66.6	73.4	79.8	80.3	72.8	63.1	51.8	36.9	
		MCDB	35.6	42.8	55.0	68.4	76.6	81.4	85.9	86.5	79.0	70.1	55.7	39.6	
5%		WB	30.4	34.6	44.9	56.2	64.7	71.8	78.0	78.3	70.9	61.0	49.1	34.1	
		MCDB	33.9	38.7	51.5	65.2	74.3	79.4	83.3	84.0	77.6	67.2	52.5	36.7	
10%		WB	27.6	32.3	41.8	53.5	62.8	70.4	76.6	77.0	69.3	58.6	46.2	31.7	
		MCDB	31.2	35.9	47.9	61.7	71.6	77.6	81.2	81.7	75.8	64.2	49.6	34.4	
Mean Daily Temperature Range		5% DB	MDBR	17.2	17.1	18.0	20.9	21.0	17.1	12.3	14.5	19.1	20.4	15.7	15.0
			MCDBR	18.9	19.8	23.8	26.8	26.7	23.3	18.3	18.3	21.9	23.5	19.7	15.6
	MCWBR		14.9	14.3	16.1	15.4	13.5	11.6	9.5	9.3	11.6	13.9	14.4	12.6	
	5% WB	MCDBR	17.5	18.1	22.3	24.5	23.2	18.6	14.3	14.3	18.0	20.9	17.5	14.5	
		MCWBR	14.5	13.9	16.2	15.4	13.2	10.6	8.7	9.0	11.0	13.7	14.1	12.6	
Clear-Sky Solar Irradiance	taub	0.394	0.471	0.534	0.615	0.641	0.716	0.689	0.584	0.511	0.537	0.478	0.407		
	taud	1.973	1.827	1.739	1.629	1.620	1.547	1.647	1.864	1.970	1.804	1.875	1.970		
	Ebn at Noon	227	224	226	218	216	200	204	223	231	201	194	209		
	Edh at Noon	41	55	66	79	81	88	79	62	52	55	45	38		
All-Sky Solar Radiation	RadAvg	776	1020	1382	1587	1787	1627	1341	1427	1405	1072	729	653		
	RadStd	45	69	99	123	148	180	161	167	115	70	78	47		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page